					Course Code:A2610001
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		ENGLISH for TECHNICAL COMMUNICATION			
		(Common to All Branches)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Identify any four expressions for asking for permissions.	**1**	**2**	**2**
	b).	Complete the sentences with the appropriate words as directed in the brackets. i. The students are interested ___ learning new technologies. (Preposition) ii. The success of _____ project depends on effective teamwork. (Article)	**2**	**3**	**2**
	c).	Construct one example sentence for any two of the following idiomatic expressions. i. A piece of cake ii. Break the ice iii. Once in a blue moon iv. Spill the beans	**3**	**3**	**2**
	d).	Write about the use of skimming technique in reading the texts.	**4**	**3**	**2**
	e).	Dos and don'ts of social media messaging.	**5**	**3**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	Develop an essay on the strategies that can help a person sound smarter, more confident, and articulate when communicating.	**1**	**3**	**5**
	b).	Write a dialogue between a student and a librarian about borrowing a reference book and renewing the library membership.	**1**	**3**	**5**
		OR			
3.		Construct a paragraph on "The Importance of Time Management" in about 100–120 words and list out the cohesive devices used in the paragraph.	**1**	**3**	**10**
		UNIT-2			
4.	**a).**	Explain how eco-conscious negotiation strategies can be applied to address an environmental issue and promote a sustainable future.	**2**	**3**	**5**

	b).	Employ appropriate verb forms to fill in the blanks. i. By next year, she ___ (finish) her degree. ii. The train ___ (arrive) at the station just now. iii. I ___ (study) English for three years. iv. When I called him, he ___ (sleep). v. Listen! Someone ___ (play) the guitar.	**2**	**3**	**5**
		OR			
5.		Apply appropriate grammatical rules and correct any **TEN** of the following sentences. i. Each of the engineers are responsible for testing the software. ii. The equipment in the laboratory were not functioning properly. iii. He discussed about the advantages of artificial intelligence. iv. The results of the experiment shows a significant improvement. v. The engineer suggested to use a different algorithm for the project. vi. If the system will fail, the backup server will be activated. vii. He is resembling his father. viii. The film show began before we reached the theatre. ix. Neither the project manager nor the team members was aware of the technical issue. x. She is an university student pursuing engineering. xi. Rahul went to school despite of having a fever. xii. I prefer chocolate milkshake than vanilla ice cream.	**2**	**3**	**10**
		UNIT-3			
6.	**a).**	Illustrate the appropriate ethical guidelines to address a challenge arising from the use of artificial intelligence or other disruptive technologies.	**3**	**3**	**5**
	b).	Construct sentences by substituting the given phrases with suitable one-word for any **FIVE** of the following. i. One who loves mankind. ii. One who cannot be defeated iii. Hobby of collecting coins. iv. Something that cannot be avoided v. A speech made without preparation vi. A person who spends money wastefully vii. A person who designs and develops software viii. A person who loves and collects	**3**	**3**	**5**
		OR			
7.	**a).**	Construct a formal letter to the Editor of a newspaper expressing your concern about the increasing use of plastic in your city. Apply appropriate letter-writing conventions and suggest possible solutions.	**3**	**3**	**10**

		UNIT-4			
8.	**a).**	Develop an essay on how innovative business practices can be applied to address challenges faced by modern startups.	**4**	**3**	**5**
	b).	Analyze the following hints to construct a formal email by applying appropriate writing techniques, tone, and structure. **Student ---> HR Manager:** New training/internship program -latest technology- useful – my career-want to be included- participate keenly-learn and implement- benefit of the company.	**4**	**3**	**5**
		OR			
9.	**a).**	Analyze the given passage and write answers to the following questions. Technology has transformed the way students learn and access information. Digital platforms, online courses, and educational applications allow learners to study at their own pace and explore subjects beyond the classroom. Engineering students, in particular, can use simulations, virtual laboratories, and collaborative tools to understand complex concepts more effectively. However, technology should complement traditional learning rather than completely replace it. Excessive dependence on digital resources may reduce face-to-face interaction and critical thinking. Therefore, students should use technology responsibly and balance digital learning with classroom discussions, practical activities, and independent study. Questions i. How has technology transformed the learning process? ii. What are the advantages of digital platforms and online courses? iii. How can engineering students benefit from simulations and virtual laboratories? iv. Why should technology not completely replace traditional learning? v. What are the possible disadvantages of excessive dependence on digital resources? vi. What does the word “complement” mean in the context of the passage? vii. Find a word from the passage that means “working together.” viii. Find a word from the passage that means “too much or more than necessary.”	**4**	**3**	**10**

		ix. What three activities does the passage recommend balancing with digital learning? x. What is the central idea of the passage?			
		UNIT-5			
10.	**a).**	Demonstrate how core values can be integrated into ethical decision-making and socially responsible leadership.	**5**	**3**	**5**
	b).	Construct the following sentences as directed in the brackets. i. She has written many novels. (Add Question Tag) ii. The report was prepared by the committee. (Convert it into Active Voice) iii. He worked hard to pass the examination. (Convert it into a Complex Sentence) iv. Very few engineering colleges in India are as reputed as IIT Madras. (Convert it into the Superlative Degree) v. Smrithi is the cleverest girl in the class. (Change it into Comparative Degree)	**5**	**3**	**5**
		OR			
11.	**a).**	Develop the content to be used in a blog on 'Social Media'.	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:A2610002

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R26**

I B.Tech. I Semester MODEL QUESTION PAPER

LINEAR ALGEBRA & CALCULUS

(Common to All Branches of Engineering)

Time: 3 Hrs. **Max. Marks: 60 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

5 x 2 = 10 Marks

			CO	KL	M
1.	(a)	Find the value of k such that rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & k & 7 \\ 3 & 6 & 10 \end{bmatrix}$ is 2	1	3	2
	(b)	Find the singular values of the matrix $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$	2	3	2
	(c)	Find the value of c in [0, 4] for $f(x) = x(x-2)(x-3)$ by using Lagrange's mean value theorem.	3	3	2
	(d)	Evaluate $\underset{\substack{x\to\infty \\ y\to 2}}{Lt}\left(\frac{xy+1}{x^2+2y^2}\right)$	4	3	2
	(e)	Evaluate $\int_0^2 \int_0^4 (x^2+y^2)\,dxdy$	5	3	2

5 x 10 = 50 Marks

		UNIT-1			
2.	(a)	Determine the rank of the matrix $\begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$ by reducing it to echelon form.	1	3	5
	(b)	Find the inverse of the matrix $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ by using Gauss – Jordan method.	1	3	5
		OR			
3.	(a)	For what values of a and b do the equations $x + 2y + 3z = 6$, $x + 3y + 5z = 9$ and $2x + 5y + az = b$ have (i) No solution (ii) An unique solution (ii) Infinitely many solutions	1	3	5
	(b)	Solve the equations $20x + y - 2z = 17$, $3x + 20y - z = -18$and $2x - 3y + 20z = 25$ by Gauss-Seidel iteration method.	1	3	5
		UNIT-2			

4	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$		2	3	10
	OR				
5.	Reduce $6x^2 + 3y^2 + 3z^2 - 4xy - 2yz + 4xz$ to sum of squares by linear transformation and hence find the matrix of the transformation, Nature, Signature and index of the quadratic form.		2	3	10
		UNIT-3			
6.	(a)	Verify Rolle's theorem for $\frac{\sin x}{e^x}$ in $(0, \pi)$.	3	3	5
	(b)	Expand $f(x) = \cos x$ by Maclaurin's theorem with Lagrange's form of remainder.	3	3	5
		OR			
7.	(a)	Find the appropriate value of c used in Cauchy's Mean-Value theorem for $f(x) = \ln x$ in [1, e]	3	3	5
	(b)	Expand $\tan^{-1}x$ in powers of $(x-1)$ upto fourth degree term.	3	3	5
		UNIT-4			
8.	(a)	If $u = \cos^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$	4	3	5
	(b)	Expand $f(x,y) = e^y \log(1+x)$ in powers of x and y using Maclaurin's series.	4	3	5
		OR			
9.	(a)	If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$	4	3	5
	(b)	Discuss the maxima and minima of $f(x,y) = x^3y^2(1-x-y)$	4	3	5
		UNIT-5			
10.	(a)	Evaluate $\int\int xydxdy$ over the positive quadrant of the circle $x^2 + y^2 = a^2$ by changing it in to polar coordinates	5	3	5
	(b)	By the change of order of integration evaluate $\int_0^{4a}\int_{x^2/4a}^{2\sqrt{ax}} dydx$.	5	3	5
		OR			
11.		Evaluate $\iiint_V \frac{dxdydz}{x^2+y^2+z^2}$ where V is the volume of the sphere $x^2 + y^2 + z^2 = a^2$	5	3	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

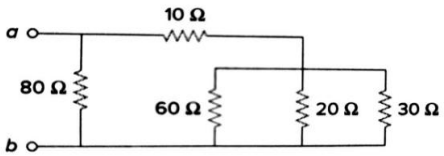
NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

				Course Code:A2610301	
		SAGI RAMAKRISHNAM RAJU ENGINEERING COLLEGE(A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		PHYSICS FOR MECHANICAL ENGINEERING			
		(For ME)			
Time:3Hrs.				**Max.Marks: 60M**	
		Answer Question No.1 compulsorily			
		Answer **ONE** Question from **EACH UNIT**			
		Assume suitable data if necessary			
				5x 2 = 10 Marks	
			CO	**KL**	**M**
1.	**a)**	Distinguish ordinary and polarized lights.	1	2	2
	b)	Define space lattice and basis.	2	2	2
	c)	Explain the physical significance of the wave function.	3	2	2
	d)	Distinguish spontaneous and stimulated emissions.	4	2	2
	e)	Mention any four properties of ultrasonics.	5	2	2
				5x 10 = 50 Marks	
		UNIT-1	**CO**	**KL**	**M**
2.		Explain the formation of Newton's rings in reflected light. Derive the mathematical expressions for the diameters of dark and bright rings.	1	3	10
		OR			
3.		Discuss the Fraunhofer diffraction at a single slit. Obtain the condition for principal maximum and minima.	1	3	10
		UNIT-2			
4.		Define coordination number and packing factor. Calculate the packing factor for SC and BCC structures.	2	3	5
		OR			
5.	**a)**	Derive an expression for the interplanar spacing between successive (h k l) plane	2	3	5
	b)	Explain how crystal structure determined by Laue method.	2	3	5
		UNIT-3			
6.		Derive the energy eigenvalues and normalized wave functions of a particle confined in a one-dimensional infinite potential well.	3	3	10
		OR			
7.	**a)**	Explain the basic principles of quantum computing .	3	3	5
	b)	Explain briefly about quantum gates.	3	3	5
		UNIT-4			
8.	**a)**	Derive the expression for the equivalent stiffness of two springs connected in	4	3	5

		series.			
	b)	Compare underdamped, critically damped, and overdamped systems.	4	4	5
		OR			
		Describe the construction and working of a He-Ne laser with a suitable energy level diagram.	4	3	10
		UNIT-5			
10.		Explain the production of ultrasonics by the piezoelectric method with a neat diagram.	5	4	10
		OR			
11.		Explain the pulse-echo method of ultrasonic Non-Destructive Testing in reflection mode with a neat block diagram	5	4	10

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2610007
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. I Semester MODEL QUESTION PAPER			
		BASIC ELECTRICAL & ELECTRONICS ENGINEERING			
		(Common to AIDS, IT & ME)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		Assume suitable data if necessary			
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	Define Ohm's law.	**1**	**1**	**2**
	b).	List the advantages of solar power plant.	**2**	**1**	**2**
	c).	Define slip.	**3**	**1**	**2**
	d).	Draw the diagram of NPN and PNP transistors.	**4**	**1**	**2**
	i).	Convert binary number to decimal number ,1011_2 to (-----)$_{10}$	**5**	**2**	**2**
					5 x 10 = 50 Marks
		UNIT-1			
2.	**a).**	State and explain Kirchhoff's Laws with example.	**1**	**3**	**5**
	b).	Calculate the equivalent resistance Rab at terminals a-b for the given circuit. 	**1**	**3**	**5**
		OR			
3.	**a).**	Derive the expression for the RMS value of a sinusoidal waveform. $v(t)=V_m \sin\omega t$.	**1**	**3**	**5**
	b).	Explain the phasor relationship between voltages and currents in a series RL circuit.	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Explain the working of thermal power plant with a neat layout.	**2**	**3**	**5**
	b).	Explain the working of CRO with neat diagram.	**2**	**3**	**5**
		OR			
5.	**a).**	Explain the working of a Digital Voltmeter (DVM) with neat diagram.	**2**	**3**	**5**
	b).	Explain working principle and operation of a transformer.	**2**	**3**	**5**
		UNIT-3			
6.	**a).**	Explain the working principle of a three-phase induction motor.	**3**	**3**	**5**
	b).	Derive the torque equation for DC Motor.	**3**	**3**	**5**

		OR			
7.	**a).**	Illustrate about the construction details and operation of a miniature circuit breaker (MCB)?	**3**	**3**	**5**
	b).	Illustrate about the construction and operation of ELCB	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain the operation of PN junction Diode and draw the V-I characteristics.	**4**	**3**	**5**
	b).	Explain the operation of transistor as a switch.	**4**	**3**	**5**
		OR			
9.	**a).**	Explain the operation of Half wave rectifier with neat sketch.	**4**	**3**	**5**
	b).	Explain how the Zener diode acts as a voltage regulator.	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Write the Truth Tables of Logic Gates – NOT, OR, AND, NOR and NAND	**5**	**2**	**5**
	b).	Convert the following number system into indicated system i) $(256)_{10}$ = (……..) $_2$ ii) $(F32C)_{16}$ = (…….) $_{10}$	**5**	**3**	**5**
		OR			
11.	**a).**	Explain and Design the Full adder circuit using half adders.	**5**	**3**	**5**
	b).	Discuss the operation of JK flip-flop.	**5**	**3**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:A2620001

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A) **R26**

I B.Tech. II Semester MODEL QUESTION PAPER

DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common to All Branches)

Time: 3 Hrs. **Max. Marks: 60 M**

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

5 x 2 = 10 Marks

			CO	KL	M
1.	**a).**	Find the integrating factor of $\frac{dy}{dx} - \frac{3y}{x} = x$	**1**	**3**	**2**
	b).	Solve$(D^2 - 4D + 4)y = 0$	**2**	**3**	**2**
	c).	Obtain a P.D.E from $z = ax + by + c$	**3**	**2**	**2**
	d).	Determine $\nabla\emptyset$, where $\emptyset = \sqrt{x^2 + y^2 + z^2}$	**4**	**3**	**2**
	e).	Find div $\bar{F}$, where $\bar{F} = x\bar{\imath} + y\bar{\jmath} + z\bar{k}$	**5**	**3**	**2**

5 x 10 =50Marks

		UNIT-1			
2.	**a).**	Solve $y \log y \, dx + (x - \log y)dy = 0$	**1**	**3**	**5**
	b).	Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$	**1**	**3**	**5**
		OR			
3.	**a).**	If the air is maintained at 30ºC and the temperature of the body cools from 100ºC to 70º C in 15 minutes, determine the time at which the temperature of the body will be 40º C.	**1**	**3**	**5**
	b).	Find the orthogonal trajectories of the family of curves $r^2 = a^2 Sin2\theta$	**1**	**3**	**5**
		UNIT-2			
4.	**a).**	Solve $(D-2)^2 y = 8(e^{2x} + x^2)$	**2**	**3**	**5**
	b).	Solve by the method of variation of parameters $\frac{d^2y}{dx^2} + y = \tan x$	**2**	**3**	**5**
		OR			
5.	**a).**	Solve simultaneous equations $\frac{dx}{dt} + y = \sin t$, $\frac{dy}{dt} + x = \cos t$, given that x = 2 and y=0 when t =0	**2**	**3**	**5**
	b).	The damped LCR circuit is governed by the differential equation $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{c} = 0$ where L, C, R are positive constants. Find the critical resistance.	**2**	**3**	**5**

		UNIT-3			
6.	**a).**	Form the partial differential equation by eliminating arbitrary function from $f(x^2+y^2+z^2, z^2-2xy)=0$.	**3**	**3**	**5**
	b).	Solve $x^2(y-z)p+y^2(z-x)q=z^2(x-y)$	**3**	**3**	**5**
		OR			
7.		Solve $\left(D^2-3DD'+2D'^2\right)z=e^{x+2y}+Sin(x+2y)$	**3**	**3**	**10**
		UNIT-4			
8.	**a).**	Find the directional derivative of $f = xy+yz+zx$ in the direction of vector $\bar{i}+2\bar{j}+2\bar{k}$ at the point (1,2,0).	**4**	**3**	**5**
	b).	If $\overrightarrow{F} = 3x^2\overrightarrow{\imath}+5xy^2\overrightarrow{j}+5xyz^3\vec{k}$,then find $\nabla(\nabla.\vec{F})$	**4**	**3**	**5**
		OR			
9.	**a).**	Prove that $(y^2-z^2+3yz-2x)\vec{\imath}+(3xz+2xy)\vec{j}+(3xy-2xz+2z)\vec{k}$ is solenoidal.	**4**	**3**	**5**
	b).	Find the angle between the surfaces $xy^2z=3x+z^2$ and $3x^2-y^2+2z=1$ at the point (1,–2,1)	**4**	**3**	**5**
		UNIT-5			
10.	**a).**	Find the work done by the force field $\vec{F}=(2x+y^2)\vec{\imath}+xz\vec{j}+xyz\vec{k}$ in moving a particle along a straight line from (0,0,0) to (2,1,3).	**5**	**3**	**5**
	b).	Evaluate $\oint_c\{(x^2y\,dx+x^2dy\}$ where C is a circle $x^2+y^2=a^2$ by using Green's theorem.	**5**	**3**	**5**
		OR			
11.		Verify Gauss – divergence theorem for $\vec{F}=2xz\vec{\imath}+yz\vec{j}+z^2\vec{k}$ where S is an upper half surface of the sphere $x^2+y^2+z^2=a^2$	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code:A2620301
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		CHEMISTRY FOR MECHANICAL ENGINEERS			
		(For ME)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 × 2 = 10 Marks
			CO	**KL**	**M**
1.	a).	Illustrate the electrochemical mechanism of pitting corrosion to explain why it occurs as a localized form of attack.	1	3	2
	b).	Apply the concept of viscosity index to select a suitable lubricant for a given operating temperature range.	2	3	2
	c).	Calculate the HCV and LCV of a given fuel sample and differentiate their significance in fuel evaluation.	3	3	2
	d).	Demonstrate the difference between thermoplastics and thermosets using suitable engineering examples.	4	3	2
	e).	Utilize the properties of graphene to identify its application in a specific mechanical engineering component.	5	3	2
					5 x 10 = 50 Marks
		UNIT-1			
2.					
	a).	Demonstrate the electrochemical theory of corrosion to describe the anodic and cathodic reactions involved in the corrosion of iron.	1	3	5
	b).	Calculate the corrosion penetration rate (CPR) of a mechanical component from the given weight-loss data and interpret its significance for material selection.	1	3	5
		OR			
3.	a).	Illustrate the principle of cathodic protection to explain how sacrificial-anode protection prevents corrosion of a steel component.	1	3	5
	b).	Use the working principle of electroplating to describe the steps involved in nickel plating of a mechanical component.	1	3	5
		UNIT-2			

4.	a).	Relate the concept of fluid-film lubrication to how a lubricant reduces friction between moving machine surfaces.	2	3	5
	b).	Interpret lubricant-property data to explain the significance of flash point and fire point in selecting a lubricant.	2	3	5
		OR			
5.	a).	Examine the concept of refractoriness under load to explain the suitability of a refractory for high-temperature service.	2	3	5
	b).	Correlate the properties of refractories with their service performance by explaining how porosity and thermal spalling influence durability.	2	3	5
		UNIT-3			
6.	a).	Sketch the construction of a hydrogen–oxygen (H2–O2) fuel cell and explain its working principle and applications in mechanical engineering.	3	3	5
	b).	Determine the calorific value of a solid or liquid fuel from bomb-calorimeter.	3	3	5
		OR			
7.	a).	Classify the major components of coal using proximate analysis and explain their significance in fuel evaluation.	3	3	5
	b).	Demonstrate how the principle of fractional distillation is used to separate useful fractions during petroleum refining.	3	3	5
		UNIT-4			
8.	a).	Utilize the principles of heat treatment to distinguish the effects of annealing, normalizing, hardening, and tempering on steel.	4	3	5
	b).	Select a suitable alloy for a specified mechanical application based on the properties of carbon steel and stainless steel.	4	3	5
		OR			
9.	a).	Categorize composite materials based on their matrix and reinforcement, and describe two engineering applications of each type.	4	3	5
	b).	Demonstrate the principle of vulcanization to explain how the properties of natural rubber are improved for engineering applications.	4	3	5
		UNIT-5			
10.	a).	Employ the CVD method to outline the synthesis of carbon nanotubes and state its role in nanomaterial preparation.	5	3	5
	b).	Examine the properties of graphene-based composites to explain their usefulness in reducing wear and friction.	5	3	5
		OR			

11.	a).	Relate the properties of high-entropy alloys to their suitability for advanced mechanical-engineering applications.	5	3	5
	b).	Illustrate the working principle of a self-healing material to explain how damage recovery occurs in an engineering component.	5	3	5

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620007
		SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)			**R26**
		I B.Tech. II Semester MODEL QUESTION PAPER			
		Computational Thinking and Problem Solving using Python			
		(Common to CE & ME)			
Time: 3 Hrs.					**Max. Marks: 60 M**
		Answer Question No.1 compulsorily			
		Answer **ONE Question** from **EACH UNIT**			
		All Questions carry equal marks			
					5 x 2 = 10 Marks
			CO	**K1**	**M**
1.	**a).**	List the five pillars of computational thinking.	**1**	**1**	**2**
	b).	Explain the Conditional Statement in Python.	**2**	**2**	**2**
	c).	Differentiate between mutable and immutable data structures.	**3**	**3**	**2**
	d).	Explain code reusability.	**4**	**2**	**2**
	e).	Define exception handling.	**5**	**1**	**2**
					5 x 10 =50Marks
		UNIT-1			
2.	**a).**	Describe the problem-solving process using decomposition, pattern recognition, abstraction, and algorithmic thinking.	**1**	**2**	**5**
	b).	Explain the standard flowchart symbols and their purposes with suitable examples.	**1**	**2**	**5**
		OR			
3.	**a).**	Explain Python variables, data types, operators, and input/output with examples.	**1**	**2**	**5**
	b).	Construct and explain a flowchart to determine whether a given number is even or odd.	**1**	**4**	**5**
		UNIT-2			
4.	**a).**	Develop a Python program to generate the multiplication table of a given number using a loop.	**2**	**4**	**5**
	b).	Explain how loops can be viewed as an application of pattern recognition in computational thinking.	**2**	**2**	**5**
		OR			
5.	**a).**	Develop a Python program to find the sum of all even numbers within a given range.	**2**	**4**	**5**
	b).	Explain how conditional statements and loops can be combined to solve a problem involving multiple cases and repetitions.	**2**	**2**	**5**

		UNIT-3			
6.	**a).**	Analyze a Python program involving lists, tuples, and dictionaries and trace the changes in the data structures.	**3**	**4**	**5**
	b).	Examine different data structures for representing key-based and ordered information in a given application.	**3**	**4**	**5**
		OR			
7.	**a).**	Explain a Python program that uses strings, lists, tuples, and dictionaries to organize and process a given dataset.	**3**	**2**	**5**
	b).	Develop a Python program to perform suitable operations on a given string.	**3**	**3**	**5**
		UNIT-4			
8.	**a).**	Explain recursion and the role of a base case in a recursive algorithm.	**4**	**2**	**5**
	b).	Explain Python modules and built-in libraries with suitable examples.	**4**	**2**	**5**
		OR			
9.	**a).**	Compare iterative and recursive approaches for solving a problem in terms of algorithmic efficiency.	**4**	**3**	**5**
	b).	Discuss responsible AI-assisted coding practices when developing Python programs.	**4**	**2**	**5**
		UNIT-5			
10.	**a).**	Develop a Python program that handles an appropriate runtime error using exception handling.	**5**	**3**	**5**
	b).	Explain the basic features and applications of NumPy, Pandas, and Matplotlib.	**5**	**2**	**5**
		OR			
11.	**a).**	Examine the role of visualization in identifying patterns in real-world datasets.	**5**	**3**	**5**
	b).	Describe how the concepts learned across all five units can be integrated into an end-to-end capstone project.	**5**	**2**	**5**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

					Course Code: A2620302
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					**R26**
I B.Tech. II Semester MODEL QUESTION PAPER					
MANUFACTURING PROCESSES					
(For ME)					
Time: 3 Hrs.					**Max. Marks: 60 M**
Answer Question No.1 compulsorily					
Answer **ONE Question** from **EACH UNIT**					
Assume suitable data if necessary					
					5 x 2 = 10 Marks
			CO	**KL**	**M**
1.	**a).**	What are the various properties of moulding sand	**1**	**1**	**1**
	b).	Explain the principle of arc welding	**2**	**2**	**1**
	c).	Explain about Tube drawing	**3**	**2**	**1**
	d).	Define blanking process	**4**	**1**	**1**
	e).	State two key advantages of plasma spraying	**5**	**1**	**1**
					5 x 10 = 50 Marks
		UNIT-1			
2.		Identify various patterns used for casting process with neat sketches.	**1**	**3**	**10**
		OR			
3.		Apply the principles of investment casting to explain the stages involved in the process	**1**	**3**	**10**
		UNIT-2			
4.		Differentiate between MIG and TIG welding.	**2**	**3**	**10**
		OR			
5.		Demonstrate any two types of resistance welding's with a neat sketches.	**2**	**3**	**10**
		UNIT-3			
6.		Apply the concept of hot rolling and explain its types	**3**	**3**	**10**
		OR			
7.		Identify the differences between hot working and cold working process.	**3**	**3**	**10**
		UNIT-4			
8.		Apply the concept of high energy rate forging by explaining it with a neat sketch and discussing its advantages.	**4**	**3**	**10**
		OR			
9.		Demonstrate Spinning, piercing and stretch forming with a neat sketch.	**4**	**3**	**10**

		UNIT-5			
10.		Apply your understanding to describe the main features of VAT photo polymerization in the additive manufacturing process.	**5**	**3**	**10**
		OR			
11.		Differentiate between PVD and CVD technologies	**5**	**3**	**10**

CO-COURSE OUTCOME **KL-KNOWLEDGE LEVEL** **M-MARKS**

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks